Course Name:	**Object Oriented Programming**	**Semester:**	**III**
Date of Performance:	**__24_ / _07__ / __2025____**	**Batch No:**	**A-4**
Faculty Name:		**Roll No:**	**16015024082**
Faculty Sign & Date:		**Grade/Marks:**	**___/15**

Experiment No: 1
Title: Basic Java Program with use of Operators and Control Flow

Aim and Objective of the Experiment:
Learn basic Java Program with use of Operators and Control Flow

COs to be achieved:
CO1: Understand concepts of Object Oriented Programming and basic characteristics of Java.

Tools used:

Theory:

(Mention about data type, operator and flow statement)

Java Scanner Class

Java **Scanner class** allows the user to take input from the console. It belongs to **java.util** package. It is used to read the input of primitive types like int, double, long, short, float, and byte. It is the easiest way to read input in a Java program.

Syntax

```
1. Scanner sc=new Scanner(System.in);
```

The above statement creates a constructor of the Scanner class having **System.in** as an argument. It means it is going to read from the standard input stream of the program. The **java.util** package should be import while using Scanner class.
It also converts the Bytes (from the input stream) into characters using the platform's default charset.

Method	Description
int nextInt()	It is used to scan the next token of the input as an integer.
float nextFloat()	It is used to scan the next token of the input as a float.
double nextDouble()	It is used to scan the next token of the input as a double.
byte nextByte()	It is used to scan the next token of the input as a byte.
String nextLine()	Advances this scanner past the current line.
boolean nextBoolean()	It is used to scan the next token of the input into a boolean value.
long nextLong()	It is used to scan the next token of the input as a long.
short nextShort()	It is used to scan the next token of the input as a Short.
BigInteger nextBigInteger()	It is used to scan the next token of the input as a BigInteger.

Code:

1. Find the perimeter and area of the circle. Input will be entered by user (You should use Math.PI constant in your program) If input is not valid then print appropriate message.
2. Consider First **n even numbers** starting from zero (0).Write a program to calculate sum of all the numbers divisible by 3 from 0 to n. Print the sum. If input is not valid then print the appropriate message.

Output:

1.

```
import java.util.Arrays;
import java.util.Scanner;

public class ExpOne {
    public static void main(String[] args) {
        System.out.println(findPerimeter());

    }
    static String findPerimeter() {
        Scanner input = new Scanner(System.in); //taking input
        int radius = input.nextInt(); //input as int
        if(radius < 0) { //edge case
            return "Invalid Input";
        }
        double pi = Math.PI; //pi
        return "Perimeter : "+  String.valueOf(2 * pi * radius).substring(0, 10) + "\nArea : " + String.valueOf(pi * Math.pow(radius, 2)).substring(0, 10);
 //returning the values with upto 10 digits
    }
```

—>

```
stella›vi ExpOne.java
stella›javac ExpOne.java
^[[A%
stella›java ExpOne
7
Perimeter : 43.9822971
Area : 153.938040
stella›
```

2.

```
import java.util.Arrays;
import java.util.Scanner;

public class ExpOne {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        int n = input.nextInt();
        System.out.println(sumOfEven(n));
    }

    static int sumOfEven(int n) {
        int count = 0; //count od prime
        int sum = 0; //sum
        int i = 0; //iteration for prime checking
        while (count < n) { //while loop for prime upto n
            if((i & 1) == 0) { //even
                if(i % 3 == 0) //div by 3
                    sum += i; //adding in sum
                count++; //incrementing even count
            }
            i++; //incrementing iterator
        }
        return sum; //returning the value
    }
```

—->

```
stella>java ExpOne
6
6
stella>java ExpOne
8
18
```

Post Lab Subjective/Objective type Questions:

1. Write a program to find the largest among three numbers x,y, and z. You should use if-then-else constructs in Java.
2. Write a program to check whether the number is an Armstrong number or not.
3. Explain the difference between class and object.
4. Why is java known as a platform independent language?

1, 2 :

```
//post labs
//1
static int maxOFThree(int a, int b, int c) {
    if(a > b && b > c) {
        return a;
    } else if (b > a && c < b) {
        return b;
    }
    return c;
}

//2
static boolean armstrong(int n) {
    String t = String.valueOf(n); //typecasting int to string for getting the length
    int ans = 0;
    int len = t.length();
    for (int i = 0; i < t.length(); i++) { //looping for each element
        ans += (int)Math.pow((t.charAt(i)-48), len); // typecasting and adding with the power of the length
    }
    return ans == n; //returning if true or false;
}
```

3) class is like a user defined data type where user can store/keep related/alike data into one data type (class). whereas an object is an instance of the class which we can use in diff classes to make use of it (perform actions on that class method).

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4) Java is known as platform independent language because of it's unique & machine independent virtual enviorment which does not ~~relate~~ relate to the physical machine and can run the byte code on its own enviorment.

Conclusion:

In this experiment, we successfully developed basic Java programs utilizing operators and control flow statements such as if-else, loops. We also used the Scanner class to accept user input and applied logic to compute results such as the perimeter and area of a circle and the sum of even numbers divisible by 3. Through this, we understood how to handle input validation and apply conditional logic to real-world problems. This experiment laid a solid foundation for understanding Java syntax, operator precedence, control flow constructs, and the concept of writing modular and interactive programs in Java.

Signature of faculty in-charge with Date: